Sahaj Bhadja

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Summary

Production-focused engineer with 2+ years supporting **Linux/Python applications, financial workflows, and distributed services**; experienced in incident triage, observability, root-cause analysis, SQL, automation, reliability, global collaboration, and performance optimization.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Monitored **production Python and SQL services** supporting 500+ users, diagnosing query and application failures through targeted checks and performance analysis while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Automated validation, reconciliation, and failure-recovery checks across 5,000+ monthly workflows, sustaining 99%+ reliability while improving concurrent processing throughput 3x and accelerating root-cause isolation for recurring operational incidents.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Supported **production Python applications** serving 150+ instructors and 10,000+ daily requests, resolving API, ingestion, and processing failures while maintaining stable service performance during peak operational periods.
- Developed multithreaded automation for secure file processing and operational reporting, handling 1,000+ files per batch while reducing feedback turnaround 60% through repeatable checks and resilient execution.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Troubleshot **cross-platform Linux application and API issues** using logs, SQL, caching, and targeted diagnostics, improving response times 30% while supporting reliable integrations across global production service environments.
- Automated ready-for-deployment checks, builds, installations, and 20+ tests through Bash, Jenkins, PyTest, and Git, generating daily failure reports for Microsoft and GitHub engineers across global Agile operations.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Monitored **eight distributed production services** with Sentry, Azure Monitor, and Kibana, diagnosing application failures across REST APIs, WebSockets, and Azure SQL in real time while securely synchronizing 2M+ records.
- Improved operational reliability through Docker packaging, Jenkins deployment pipelines, automated testing, caching, and process optimization, achieving 95%+ build validation while reducing deployment time 50% and resource waste 42%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Supported **production financial transaction applications** spanning pricing, payments, refunds, disputes, and reporting, securely processing 50,000+ records through Flask and Firebase services while sustaining sub-200-millisecond API responses.
- Performed detailed post-transaction validation across payment statuses, webhooks, refunds, and real-time updates, documenting 12+ integration flows while automating manual exception handling 70% and improving operational reporting accuracy.

Skills

Operations: Incident management, root-cause analysis, outage triage, service readiness checks, transaction reconciliation, production support, process improvement
Systems: Linux/Unix, Bash, Python, TCP/IP, networking, concurrency, distributed systems, high availability, performance optimization
Observability: Sentry, Azure Monitor, Kibana, logging, monitoring, alert investigation, system diagnostics, operational reporting
Data & Markets: PostgreSQL, MSSQL, SQL, Firebase, transactions, pricing, payments, refunds, financial applications
Delivery: Automation scripts, Docker, Kubernetes, AWS, Azure, Git, Jenkins, GitHub Actions, CI/CD, testing, global collaboration

Projects

FinNexus Live | *Kubernetes, MongoDB, WebSockets*

- Built a **real-time financial application** supporting 50,000+ users and 1M+ transactions with secure bank integrations, live status updates, advanced filtering, MongoDB persistence, and Kubernetes-based production deployment automation.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, MSSQL*

- Enhanced a **cross-platform database application** supporting PostgreSQL, MSSQL, and 8+ additional systems with schema exploration, query generation, Docker, Maven, Linux workflows, validation, and reusable structured debugging interfaces.

CPU Scheduling Simulator Live | *C++, React.js, Operating Systems*

- Implemented and benchmarked **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case handling, and dynamic performance metrics across 1,000+ process combinations, reducing average turnaround time 25%.